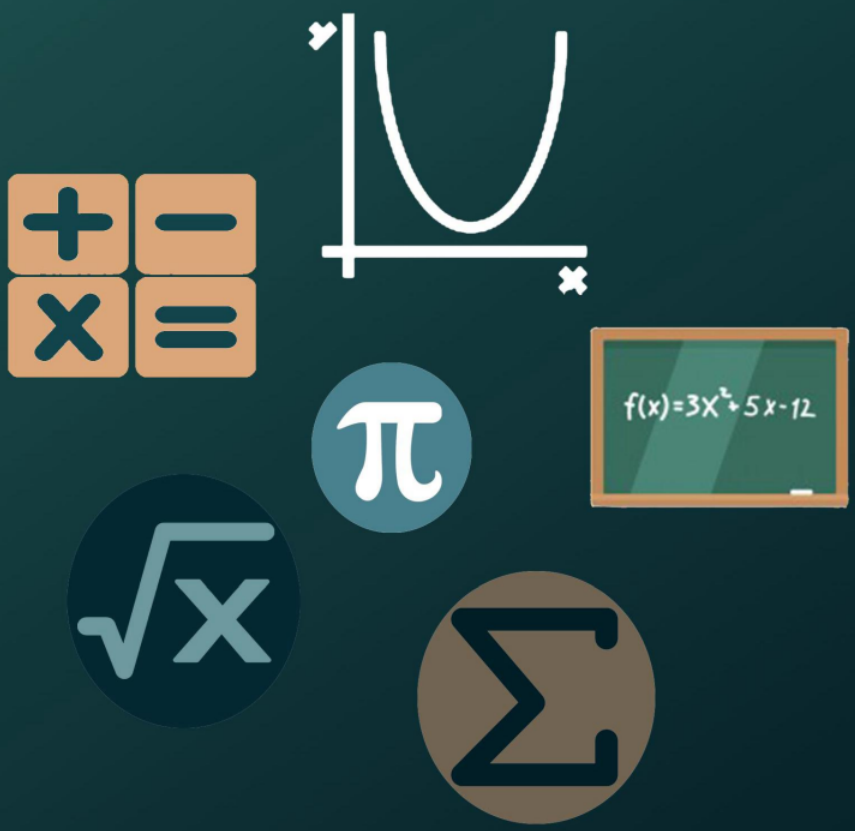
MATHEMATICS

Point



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex-1	Ex-1A	Ex-2	Ex-2A
1	A Mathematical Contribution of a Fly				
2	Coordinate System & Distance Formula	1 - 5	1, 2, 8, 14	1	8, 13
3	Section Formula	6 - 8	3, 4, 15,		1, 9
4	Centers of a Triangle - Centroid, Incentre & Excentre	9 - 12	16		2, 14-15
5	Centers of a Triangle - Orthocentre & Circumcentre	13,14	5, 6	2	10, 16
6	Area of Triangle		7, 11 - 13, 17 - 18	3, 4	11, 17
7	Locus	15 - 18	9, 19, 20	5	3, 5-7, 18-21
8	After Complete Chapter		10	6 - 8	4, 12, 22-23

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Coordinate System & Distance Formula

Q1. Coordinates of the vertices of a triangle ABC are (12,8), (–2,6) and (6,0) then the **correct** statement is-

(A) triangle is right but not isosceles

(B) triangle is isosceles but not right

(C) triangle is obtuse

(D) the product of the abscissa of the centroid, orthocenter and circumcenter is 160.

Q2. Find the value of x_1 if the distance between the points $(x_1, 2)$ and (3, 4) be 8.

(A) $3 \pm 2\sqrt{15}$ (B) $3 \pm \sqrt{15}$

(C) $2 \pm 3\sqrt{15}$ (D) $2 \pm \sqrt{15}$

Q3. If P(1,2), Q(4,6), R(5,7) & S(a,b) are the vertices of a parallelogram PQRS, then :

(A) a = 2, b = 4 (B) a = 3, b = 4

(C) a = 2, b = 3 (D) a = 3, b = 5

Q4. The four points whose co-ordinates are (2,1),(1,4),(4,5),(5,2) form :

(A) A rectangle which is not a square

(B) A trapezium which is not a parallelogram

(C) A square

(D) A rhombus which is not a square

Q5. The length of a line segment AB is 10 units. If the coordinates of one extremity are (2,–3) and the abscissa of the other extremity is 10 then the sum of all possible values of the ordinate of the other extremity is -

(A) 3 (B) –4 (C) 12 (D) –6

Section Formula

Q6. The length of median through A of a triangle ABC whose vertices are A(–1, 3), B(1, –1) & C(5, 1)**:**

(A) $5\sqrt{2}$ unit (B) 5 unit

(C) $\sqrt{5}$ unit (D) $\frac{5}{2}$ unit

Q7. The coordinates of a point which divides externally the line joining (1, –3) & (–3, 9) in the ratio 1 : 3 :

(A) (3, 9) (B) (3, –9)

(C) (–3, 9) (D) (–3, –9)

Q8. If A and B are the points (–3,4) and (2,1), then the co-ordinates of the point C on AB produced such that AC = 2BC are :

(A) (2,4) (B) (3,7)

(C) (7,–2) (D) $\left(-\frac{1}{2}, \frac{5}{2}\right)$

Centers of a Triangle - Centroid, Incentre & Excentre

Q9. Two vertices of a triangle are (–2, 5) and (6, 3). If its centroid is (1, 4). Then third vertex is **:**

(A) (1, 4)

(B) (–1, 4)

(C) (–1, –4)

(D) (1, –4)

Q10. The medians of a triangle meet at (0,–3) and its two vertices are at (–1,4) and (5,2). Then the third vertex is at -

(A) (4,15)

(B) (–4,–15)

(C) (–4,15)

(D) (4,–15)

Q11. The incentre of the triangle with vertices $(1, \sqrt{3})$, (0, 0) and (2, 0) is -

(A) $\left(1, -\frac{\sqrt{3}}{2}\right)$ (B) $\left(\frac{2}{3}, \frac{1}{\sqrt{3}}\right)$

(C) $\left(1, \frac{1}{\sqrt{3}}\right)$ (D) None

Q12. Consider the points P(2,–4); Q(4,–2) and R(7,1). The points P,Q,R -

(A) form an equilateral triangle

(B) form a right angled triangle

(C) form an isosceles triangle which is not equilateral

(D) are collinear.

Centers of a Triangle - Orthocentre & Circumcentre

Q13. If in triangle ABC, A ≡ (1,10), circumcenter $\equiv \left(-\frac{1}{3}, \frac{2}{3}\right)$ and orthocenter $\equiv \left(\frac{11}{3}, \frac{4}{3}\right)$ then the co-ordinates of mid-point of side opposite to A is-

(A) (1,–11/3) (B) (1,5)

(C) (1,–3) (D) (1,6)

Q14. The orthocenter of the triangle ABC is 'B' and the circumcenter is 'S' (a,b). If A is the origin then the co-ordinates of C are :

(A) (2a,2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$

(C) $\left(\sqrt{a^2+b^2}, 0\right)$ (D) None

Locus

Q15. A point P(x,y) moves so that the sum of the distance from P to the coordinate axes is equal to the distance from P to the point A(1,1). The equation of the locus of P in the first quadrant is -

(A) (x + 1) (y + 1) = 1

(B) (x + 1) (y + 1) = 2

(C) (x – 1)(y – 1) = 1

(D) (x – 1)(y – 1) = 2

Q16. Let A(2,–3) and B(–2,1) be vertices of a ΔABC. If the centroid of ΔABC moves on the line 2x + 3y =1, then the locus of the vertex C is-

(A) 2x + 3y = 9

(B) 2x – 3y = 7

(C) 3x + 2y = 5

(D) 3x – 2y = 3

Q17. Given the points A(0,4) and B(0,–4), the equation of the locus of the point P such that |AP – BP| = 6 is-

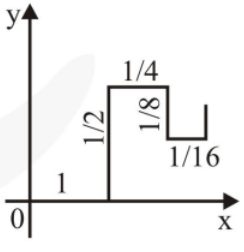
(A) $9x^2 - 7y^2 + 63 = 0$

(B) $9x^2 - 7y^2 - 63 = 0$

(C) $7x^2 - 9y^2 + 63 = 0$

(D) $7x^2 - 9y^2 - 63 = 0$

Q18. A particle begins at the origin and moves successively in the following manner as shown, 1 unit to the right, 1/2 unit up, 1/4 unit to the right, 1/8 unit down, 1/16 unit to the right etc.



The length of each move is half the length of the previous move and movement continues in the 'zigzag' manner infinitely. The co-ordinates of the point to which the 'zigzag' converges is -

(A) (4/3, 2/3) (B) (4/3,2/5)

(C) (3/2,2/3) (D) (2,2/5)

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. A line is of length 10 and one end is at the point (2, –3) if the abscissa of the other end is 10, then its ordinate is :-

(A) 9 (B) –3

(C) 3 (D) –9

Q2. If vertices of a quadrilateral are A(0, 0), B(3, 4), C(7, 7), D(4, 3), then :-

(A) Quadrilateral ABCD is a Square

(B) Quadrilateral ABCD is a rhombus

(C) Area of ABCD is 25 square units

(D) Area of ABCD is 7 square units

Q3. Which of the following are correct statements :-

(A) Point $C\left(1,\frac{4}{5}\right)$ divides the line joining (–1, 2) & (4, –5) in the ratio 2 : 3 internally.

(B) $\left(\frac{10}{7},\frac{33}{7}\right)$ divides the line joining (1, 3) & (2, 7) in the ratio 3 : 4 internally.

(C) (–11, 16) divides (–1, 2) & (4, –5) in the ratio 2 : 3 externally.

(D) (–2, –9) divides (1, 3) & (2, 7) in the ratio 3 : 4 externally.

Q4. The point A(–sinθ, cosθ) is $\sqrt{7}$ units away from the point B(2cos75°, 2sin75°). If $\theta\in[0, 2\pi)$ then possible value of θ are :

(A) 105° (B) 85°

(C) 45° (D) 225°

Q5. If A(2, 3), B(0, –1) & C(4, 2) are vertices of a triangle then :-

(A) Co-ordinates of centroid are $\left(2,\frac{4}{3}\right)$

(B) Co-ordinates of circumcentre are $\left(2,\frac{3}{2}\right)$

(C) Co-ordinates of orthocentre are (2, 3)

(D) ABC is an isosceles triangle

Q6. For a triangle ΔPQR P(1, 10) circumcentre $\equiv\left(-\frac{1}{2},\frac{2}{3}\right)$ orthocentre is $\left(\frac{11}{3},\frac{4}{3}\right)$ then :-

(A) co-ordinates of centroid are $\left(\frac{8}{9},\frac{8}{9}\right)$

(B) mid point of QR is $\left(\frac{5}{6},\frac{-11}{3}\right)$

(C) co-ordinates of centroid are $\left(\frac{41}{6},\frac{10}{9}\right)$

(D) mid point of QR is $\left(\frac{39}{4},\frac{-10}{3}\right)$

Q7. If the area of the triangle with vertices (x, 0), (1, 1) and (0, 2) is 4 square units then value of x is:-

(A) –4 (B) –6

(C) 10 (D) 8

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q8.

Column-I	**Column-II**
(A) The points (2,–2), (8,4), (5,7) and (–1,1) taken in order constitute the vertices of a	(P) square
(B) The points (0,–1),(2,1),(0,3) and (–2,1) taken in order are the vertices of a	(Q) rectangle
(C) The points (3,–5), (–5,–4),(7,10),(15,9) taken in order are the vertices of a	(R) trapezium
(D) The points (–3,4), (–1,0), (1,0) and (3,4) taken in order are the vertices of a	(S) parallelogram
	(T) cyclic quadrilateral

MATCHING LIST TYPE

*Each question has TWO (02) **matching lists: LIST I and LIST II**. FOUR options are given representing matching of elements from LIST I and LIST II. ONLY ONE of these four options corresponds to a correct matching. For each question, choose the option corresponding to the correct matching.*

Q9. Let P(x, y) be any point on the locus then observe the following lists

Column-I	**Column-II**
(P) The sum of the squares of distance from the coordinate axis is 25	**(i)** $x^2 + y^2 = 25$
(Q) distances from x - axis and y - axis are in the ratio 2 : 3 respectively	**(ii)** $4x^2 - 9y^2 = 0$
(R) Such that square of it's distance from origin is 4 time its y-coordinate	**(iii)** $x^2 + y^2 = 4y$
(S) Distance of P from (4, 0) is double the distance of P from x-axis	**(iv)** $x^2 - 3y^2 - 8x + 16 = 0$
	(v) $9x^2 + 4y^2 = 0$

The correct matching is -

(A) (a)-i (b)-ii (c)-iii (d)-iv

(B) (a)-i (b)-v (c)-iii (d)-iv

(C) (a)-i (b)-ii (c)-iv (d)-iii

(D) (a)-ii (b)-i (c)-iii (d)-v

TRIPLE MATCH COLUMN TYPE

*This section contains ONE table having 3 columns and 4 rows. Based on the table each question has FOUR options [A], [B], [C] AND [D]. **ONLY ONE** of these four is correct.*

Q10.

Column-I Vertices of triangle	Column-II Centroid of triangle	Column-III Area of triangle (square units)
(1) (1, 2), (4, 6), (4, 2)	(p) (1, 1)	(a) 18
(2) (–1, 0), (1, 0), (3, 4)	(q) (5, 3)	(b) 4
(3) (0, –1), (2, 1), (1, 3)	(r) $\left(3,\frac{10}{3}\right)$	(c) $\frac{7}{2}$
(4) (2, –2), (8, 4), (5, 7)	(s) $\left(1,\frac{4}{3}\right)$	(d) 6

The correct matching is -

(A) (1) → (q) → (a)

(B) (2) → (s) → (b)

(C) (3) → (r) → (d)

(D) (4) → (p) → (c)

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q11. If mid points of sides of a triangle are (0, 4), (3, 6) & (–8, –2) then area of the triangle is.

Q12. If points (5, 5), (10, k) & (–5, 1) are collinear then value of k is.

Q13. Find the area of the pentagon whose vertices taken in order are (0,4),(3,0),(6,1),(7,5) and (4,9).

SUBJECTIVE TYPE

Q14. Find the distance between the following pairs of points

(i) (4, –7) and (–1, 5)

(ii) (a, 0) and (0, b)

(iii) (b + c, c + a) and (c + a, a + b)

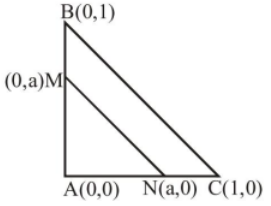
(iv) (a cosα, a sinα) and (a cosβ, a sinβ)

(v) (am_1^2, $2am_1$) and (am_2^2, $2am_2$)

Q15. The line joining the points (1,–2) and (–3,4) is trisected, find the coordinates of the points of the trisection.

Q16. If the two vertices of a triangle are (7,2) and (1,6) and its centroid is (4,6) then the coordinate of the third vertex are (a,b). The value of (a + b), is -

Q17. ΔABC lies in the plane with A(0,0), B(0,1) and C(1,0). Points M and N are chosen on AB and AC, respectively, such that MN is parallel to BC and MN divides the area of ΔABC in half. Find the coordinates of M.



Q18. Find the areas of the quadrilaterals the coordinates of whose angular points, taken in order, are
(i) (1,1), (3, 4) (5, –2) and (4, –7)
(ii) (–1, 6) (–3, –9), (5, –8) and (3, 9)

Q19. Find the equation to the locus of a point which moves so that
(i) Its distance from the axis of x is three times its distance from the axis of y.
(ii) Its distance from the point (a, 0) is always four times its distance from the axis of y.
(iii) The sum of the squares of its distances from the axes is equal to 3.

Q20. Find the locus of a point which is equidistant from the points (3,4) and (5,–2).

Exercise - 2 — *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If the point (x, y) be eqniststant from the point (4, 2) and (–3, –5). Then :
(A) $x + y - 1 = 0$ (B) $x - y - 1 = 0$
(C) $x + y + 1 = 0$ (D) $x - y + 1 = 0$

Q2. The orthocentre of a ΔABC is 'B' and circumcentre is S(a, b). If A is origin then coordinate of C is-
(A) (2a, 2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$
(C) $(\sqrt{a^2+b^2}, 0)$ (D) None of these

Q3. If A(6, 3) B(–3,5), C(4–2) and D(x, 3x) are four points If the ratio of area of ΔDBC and ΔABC is 1 : 2, then the value of x, will be :-
(A) 11/8 (B) 8/11
(C) 3 (D) None of these

Q4. The line x = c cuts the triangle with corners (0,0); (1,1) and (9,1) into two region. For the area of the two regions to be the same c must be equal to-
(A) 5/2 (B) 3
(C) 7/2 (D) 3 or 15

Q5. A stick of length 10 units rests against the floor and a wall of a room. If the stick begins to slide on the floor then the locus of its middle point is :
(A) $x^2 + y^2 = 2.5$ (B) $x^2 + y^2 = 25$
(C) $x^2 + y^2 = 100$ (D) None

Q6. The harmonic conjugates of the point R(5, 1) w.r.t. the points A(2, 10) & B(6, –2) is :
(A) (8, –8) (B) (–8, 8)
(C) (–8, –8) (D) (8, 8)

Q7. Suppose ABC is a triangle with 3 acute angles A,B and C. The point whose coordinates are (cosB – sinA, sinB – cosA) can be in the -
(A) First and 2nd quadrant
(B) Second the 3rd quadrant
(C) Third and 4th quadrant
(D) Second quadrant only

Q8. A triangle has two of its vertices at (0,1) and (2,2) in the cartesian plane. Its third vertex lies on the x-axis. If the area of the triangle is 2 square units then the sum of the possible abscissae of the third vertex, is-
(A) –4 (B) 0
(C) 5 (D) 6

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If the points P(1, 3), Q(4, 7) and R are collinear. If PR = 7.5 then possible coordinates of R are :-

(A) $\left(\frac{11}{2},9\right)$ (B) $\left(\frac{17}{2},18\right)$

(C) $\left(\frac{13}{2},12\right)$ (D) $\left(-\frac{7}{2},-3\right)$

Q2. If (α, β) be the coordinates of incentre of ΔABC whose vertices are A(4, –2), B(–2, 4) & C(5, 5) then :-

(A) $2\alpha + 10\beta = 30$ (B) $\alpha + \beta = 5$

(C) $3\alpha - \beta = 5$ (D) $-5\alpha + 3\beta = -5$

Q3. If A(a, 0) & B(–a, 0) are fixed points then which of the following statements are correct :-

(A) Locus of P such that $PA^2 - PB^2 = 8$ is $ax + 2 = 0$

(B) Locus of P such that $PA = 3PB$ is $2(x^2 + y^2 + a^2) + 5ax = 0$

(C) Locus of P such that $PA^2 - PB^2 = 8$ is $2ax + 1 = 0$

(D) Locus of P such that $PA = 3PB$ is $5(x^2 + y^2 + a^2) + 4ax = 0$

Q4. The coordinates of the points A and B are respectively (–3, 2) and (2, 3). P and Q are points on the line joining A and B such that AP = PQ = QB. A square PQRS is constructed on PQ as one side, the coordinates of R can be

(A) $\left(-\frac{4}{3},\frac{7}{3}\right)$ (B) $\left(0,\frac{13}{3}\right)$

(C) $\left(\frac{1}{3},\frac{8}{3}\right)$ (D) $\left(\frac{2}{3},1\right)$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 5 to 7)

A locus is the curve traced out by a point which moves under certain geometrical conditions:

To find the locus of a point first we assume the co-ordinates of the moving point as (h,k) and then try to find a relation between h and k with the help of the given conditions of the problem. If there is any variable involved in the process then we eliminate them. At last we replace h by x and k by y and get the locus of the point which will be an equation in x and y.

Q5. Locus of centroid of the triangle whose vertices are (acost, asint), (bsint, – bcost) and (1, 0) where t is a parameter is -

(A) $(3x - 1)^2 + (3y)^2 = a^2 - b^2$

(B) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$

(C) $(3x + 1)^2 + (3y)^2 = a^2 + b^2$

(D) $(3x + 1)^2 + (3y)^2 = a^2 - b^2$

Q6. A variable line cuts x-axis at A, y-axis at B where OA = a, OB = b (O as origin) such that $a^2 + b^2 = 1$ then the locus of circumcentre of Δ OAB is -

(A) $x^2 + y^2 = 4$ (B) $x^2 + y^2 = 1/4$

(C) $x^2 - y^2 = 4$ (D) $x^2 - y^2 = 1/4$

Q7. The locus of the point of intersection of the lines $x\cos\alpha + y\sin\alpha = a$ and $x\sin\alpha - y\cos\alpha = b$ where α is variable is -

(A) $x^2 + y^2 = a^2 + b^2$ (B) $x^2 + y^2 = a^2 - b^2$

(C) $x^2 - y^2 = a^2 - b^2$ (D) $x^2 - y^2 = a^2 + b^2$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q8.

Column-I	**Column-II**
(A) The points (2,–2), (–2,1) and (5,2)	(P) Are the vertices of a right angled triangle
(B) The points (1,–2),(–3,0) and (5,6)	(Q) Are the vertices of a right angle isosceles triangle
(C) The points (3,7),(6,5) and (15,–1)	(R) Are the vertices of an equilateral triangle
(D) The points (2,2), (–2,–2) and $\left(-2\sqrt{3}, 2\sqrt{3}\right)$	(S) Do not form a triangle

Q9.

Column-I	**Column-II**
(A) The ratio in which the point $\left(\frac{10}{7}, \frac{33}{7}\right)$ divides the line segment joining the points (1, 3) and (2, 7) is p : q, then (p + q) is (p & q are coprime)	(P) $\frac{1}{5}$
(B) The ratio in which the point (–2, –9) divides the line segment joining the points (1, 3) and (2, 7) is p : q, then (p + q) is (p & q are coprime)	(Q) –1
(C) If (p, q) divides internally the line joining (–1, 2) and (4, –5) in ratio 2 : 3 then (p + q) is	(R) 5
(D) If (p, q) divides externally the line joining (–1, 2) (4, –5) in ratio 2 : 3 then (p + q) is	(S) 7

Q10. **Column-I** **Column-II**

(A) Let 'P' be a point inside the ΔABC and is equidistant from its sides. DEF is a triangle obtained by the intersection of the external angle bisectors of the angles of ΔABC. With respect to ΔDEF point P is. (P) Centroid

(B) Let 'Q' be a point inside the ΔABC if $(AQ)\times\sin\frac{A}{2}=(BQ)\times\sin\frac{B}{2}=(CQ)\times\sin\frac{C}{2}$ then with respect to ΔABC, Q is its. (Q) Orthocentre

(C) Let 'S' be a point in the ΔABC. If the point is such that infinite normals can be drawn from it on the circle passing through A, B and C then with respect to the triangle ABC, S is its. (R) Incentre

(D) Let ABC be a triangle. D is some point on the side BC such that the line segments parallel to BC with their extremities on AB and AC get bisected by AD. Point E and F are similarly obtained on CA and AB. If segments AD, BE, and CF are concurrent at a point R then with respect to the triangle ABC, R is its. (S) Circumcentre

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q11. Consider the triangle with vertices A(–2,4), B(10,–2), C(–2,–8). If G is the centroid of the triangle, find the area of the triangle BGC.

Q12. A triangle ABC is formed by three lines $x+y+2=0$, $x-2y+5=0$ and $7x+y-10=0$. P is a point inside the triangle ABC such that areas of the triangles PAB, PBC and PCA are equal. If the co-ordinates of the point P are (a,b) and the area of the triangle ABC is δ, then find $(a+b+\delta)$.

SUBJECTIVE TYPE

Q13. If a and b are real numbers between 0 and 1 such that the point (a,1), (1,b) and (0,0) form an equilateral triangle, find :

(1) a and b

(2) 2(a + b)– ab

Q14. In ΔABC it's medians AD & BE are perpendicular and AC = 4, BC = 3. If length of side AB is $\sqrt{P}$ then value of P is :

Q15. If α, β, γ are roots of the cubic $x^3-6x^2+12x-1=0$. A triangle is made by taking $\left(\alpha,\frac{1}{\alpha}\right)$, $\left(\beta,\frac{1}{\beta}\right)$ & $\left(\gamma,\frac{1}{\gamma}\right)$ as vertices. If centroid of the triangle is (a, b) then value of a + b is :

Q16. (1) Find the circumcenter and circumradius of the triangle whose vertices are $(1,1), (2,-1), (3, 2)$

(2) Find the distances between the points $\left(t_1^2, 2t_1\right)$ and $\left(t_2^2, 2t_2\right)$ if t_1 and t_2 are the roots of $x^2 - 2\sqrt{3}x + 2 = 0$.

Q17. Find the areas of the triangle the coordinates of whose angular points are respectively

(1) $(a\cos\phi_1, b\sin\phi_1)$, $(a\cos\phi_2, b\sin\phi_2)$ and $(a\cos\phi_3, b\sin\phi_3)$

(2) $(am_1^2, 2am_1)$, $(am_2^2, 2am_2)$ and $(am_3^2, 2am_3)$

(3) $\left(am_1, \frac{a}{m_1}\right)$, $\left(am_2, \frac{a}{m_2}\right)$ and $\left\{am_3, \frac{a}{m_3}\right\}$

Q18. Find the equation to the locus of a point which moves so that

(1) The square of its distance from the point (0, 2) is equal to 4.

(2) Its distance from the point (3, 0) is three times its distance from (0, 2).

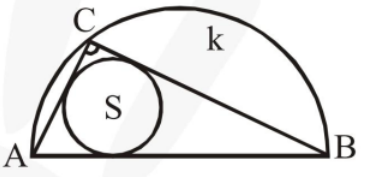
(3) Its distance from the axis of x is always one half is distance from the origin.

Q19. The ends of the hypotenuse of a right angled triangle are (6,0) and (0,6). Find the locus of the third vertex and interpret the locii geometrically.

Q20. If the distances of a point P from the points (2,1) and (1,2) are in the ratio 2 : 1, find the locus of the point P.

Q21. A variable line passes through P(2,3) and cuts the co-ordinates axes at A and B. If the parallelogram OACB (where O is the origin) is completed then find number of ordered pairs (x,y) of integers which lie on the locus of point C.

Q22. AB is the diameter of a semicircle k, C is an arbitrary point on the semicircle (other than A or B) and S is the centre of the circle inscribed into triangle ABC, then prove that measure of angle ASB = 135° for all C.



Q23. If O be the origin, and if the coordinates of any two points P_1 and P_2 be respectively (x_1,y_1) and (x_2,y_2), prove that $OP_1.OP_2 \cos P_1OP_2 = x_1x_2 + y_1y_2$.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	A	C	C	D	B	B	C	B	B
11	12	13	14	15	16	17	18		
C	D	A	A	B	A	A	B		

Ex - 1A

1. (C,D) **2.** (B,D) **3.** (B,C,D) **4.** (A,D) **5.** (A, C)

6. (A,B) **7.** (B, C)

8. (A)→(Q,S,T); (B)→(P,Q,S,T); (C)→(S); (D)→(R,T)

9. (A) **10.** (B) **11.** 4 **12.** 7 **13.** 36.5

14. **(i)** 13 **(ii)** $\sqrt{a^2+b^2}$ **(iii)** $\sqrt{a^2+2b^2+c^2-2ab-2bc}$ **(iv)** $\left|2a\sin\frac{(\alpha-\beta)}{2}\right|$

(v) $|a(m_1-m_2)\sqrt{(m_1+m_2)^2+4}|$

15. $\left(\frac{-1}{3},0\right);\left(\frac{-5}{3},2\right)$ **16.** 14 **17.** $\left(0,\frac{1}{\sqrt{2}}\right)$

18. **(i)** $20\frac{1}{2}$ **(ii)** 96 **19.** **(i)** $y = 3x$, **(ii)** $15x^2 - y^2 + 2ax = a^2$, **(iii)** $x^2 + y^2 = 3$

20. $x - 3y = 1$

Ex-2							
1	2	3	4	5	6	7	8
C	A	A	B	B	A	D	A

Ex - 2A

1. (A,D) **2.** (A,B,C,D) **3.** (A,B) **4.** (B,D) **5.** (B)

6. (B) **7.** (A)

8. (A)→(P,Q); (B)→(P); (C)→(S); (D)→(R)

9. (A)→S; (B)→S; (C)→P; (D)→R

10. A→Q, B→R , C→S, D→P **11.** 24 **12.** 15

13. **(1)** $a = b = 2 - \sqrt{3}$ **(2)** 1 **14.** 5 **15.** 6

16. **(1)** $\left(\frac{5}{2}, \frac{1}{2}\right); \sqrt{\frac{5}{2}}$ **(2)** 8

17. **(1)** $\left|2ab\sin\frac{\phi_2-\phi_3}{2}\sin\frac{\phi_3-\phi_1}{2}\sin\frac{\phi_1-\phi_2}{2}\right|$

(2) $a^2(m_2 - m_3)(m_3 - m_1)(m_1 - m_2)$

(3) $\frac{1}{2}a^2(m_2 - m_3)(m_3 - m_1)(m_1 - m_2) \div m_1m_2m_3$

18. **(1)** $x^2 + y^2 = 4y$

(2) $8x^2 + 8y^2 + 6x - 36y + 27 = 0$

(3) $x^2 = 3y^2$

19. $x^2 + y^2 - 6x - 6y = 0$

20. $3x^2 + 3y^2 - 4x - 14y + 15 = 0$

21. 7

-: Important Notes & Formulas :-